



Faculty of Technology

New Generation Engineering...

BLM 2010

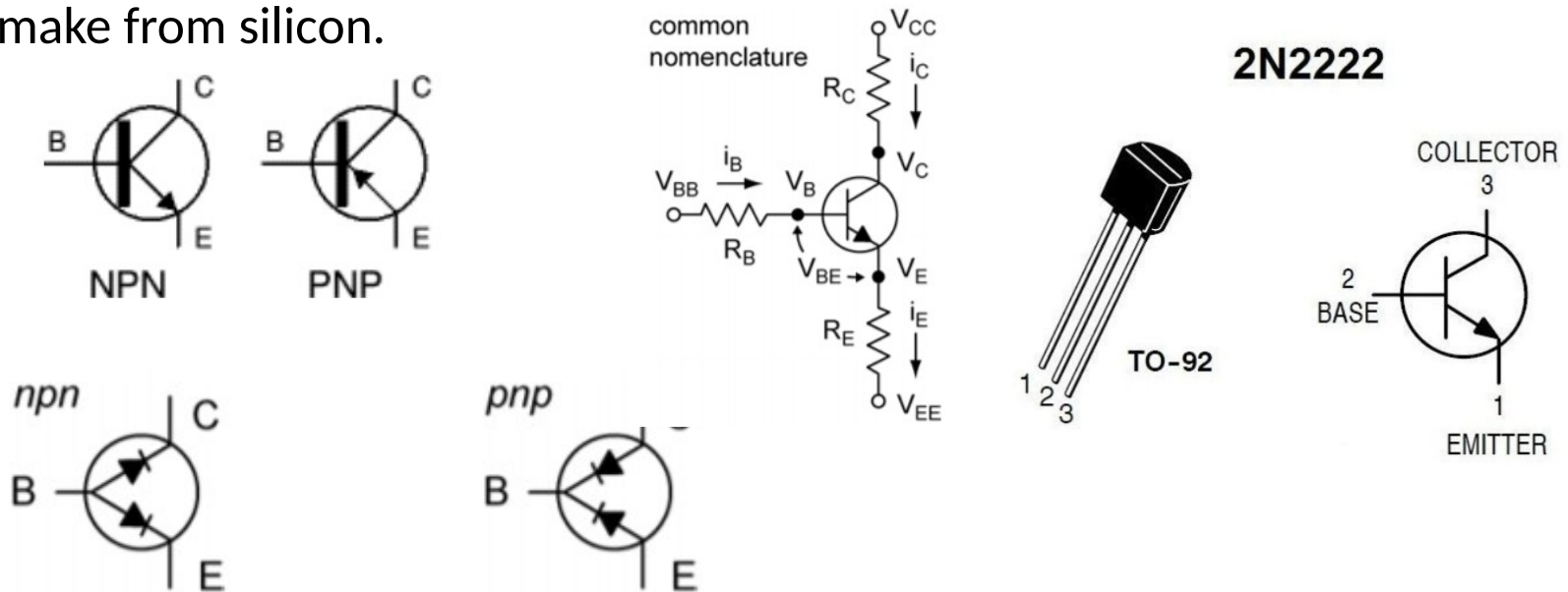
Elektronik Devrelere Giriş

Transistörlü Devreler

Dr. Öğr. Üyesi Timur İNAN

BJT

- Most transistors used today are NPN because this is the easiest type to make from silicon.



The leads are labelled **base** (B), **collector** (C) and **emitter** (E).

$$(V_{BE} = 0.7 \text{ Volt}) \quad \beta = \frac{I_C}{I_B} \quad I_E = I_C + I_B \quad I_E = (\beta + 1)I_B$$

Gain

- Gain is defined as the ratio of the output signal to the input signal.

General definition : $A = \frac{d(Output)}{d(Input)}$ if $(Output) = 0$ when $(Input) = 0$, then $A = \frac{(Output)}{(Input)}$

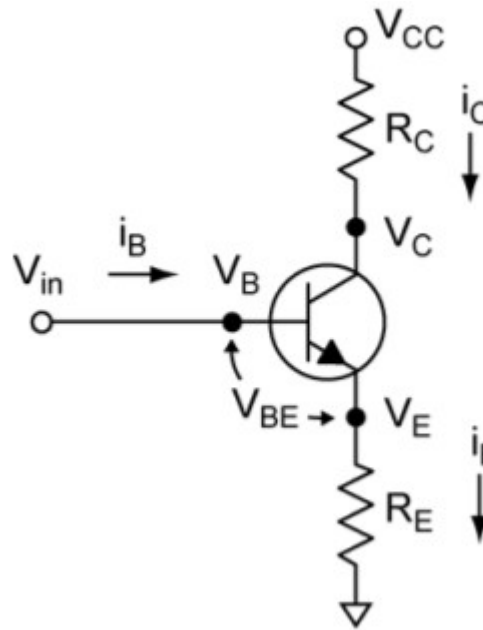
Voltage Gain : $A_v = \frac{dV_{out}}{dV_{in}}$ if $V_{out} = 0$ when $V_{in} = 0$, then $A = \frac{V_{out}}{V_{in}}$

Current Gain : $A_i = \frac{dI_{out}}{dI_{in}}$ if $I_{out} = 0$ when $I_{in} = 0$, then $A = \frac{I_{out}}{I_{in}}$

Power Gain : $A_p = \frac{dP_{out}}{dP_{in}}$ if $P_{out} = 0$ when $P_{in} = 0$, then $A = \frac{P_{out}}{P_{in}}$

Input Impedance

$$Z_{in} = \frac{dV_{in}}{dI_{in}}$$



$$V_{in} = V_B = V_{BE} + V_E = V_{BE} + I_E R_E$$

$$I_E = I_C + I_B = \beta I_B + I_B = I_B (\beta + 1)$$

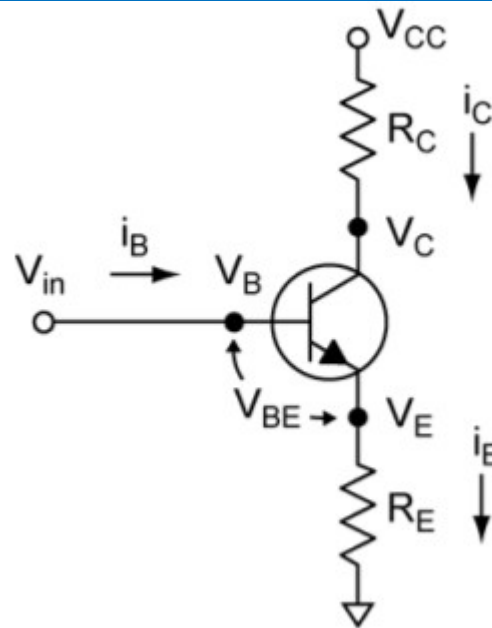
$$V_{in} = V_{BE} + I_E R_E = V_{BE} + I_B (\beta + 1) R_E$$

$$I_B = I_{in}$$

$$V_{in} = V_{BE} + I_{in} (\beta + 1) R_E$$

http://www.nhn.ou.edu/~bumm/ELAB/Lect_Notes/BJT_FET_transitors_v1_1

Input Impedance



$$Z_{in} = \frac{dV_{in}}{dI_{in}} = \frac{\alpha}{dI_{in}} (V_{BE} + I_{in}(\beta + 1)R_E) = (\beta + 1)R_E$$

$$\boxed{Z_{in} = (\beta + 1)R_E \approx \beta R_E}$$

Output Impedance

$$Z_{out} = -\frac{dV_{out}}{dI_{out}}$$

$$V_{in} = I_B R_S + V_{BE} + V_E$$

$$V_{out} = V_E$$

$$V_{out} = V_E = V_{in} - I_B R_S - V_{BE}$$

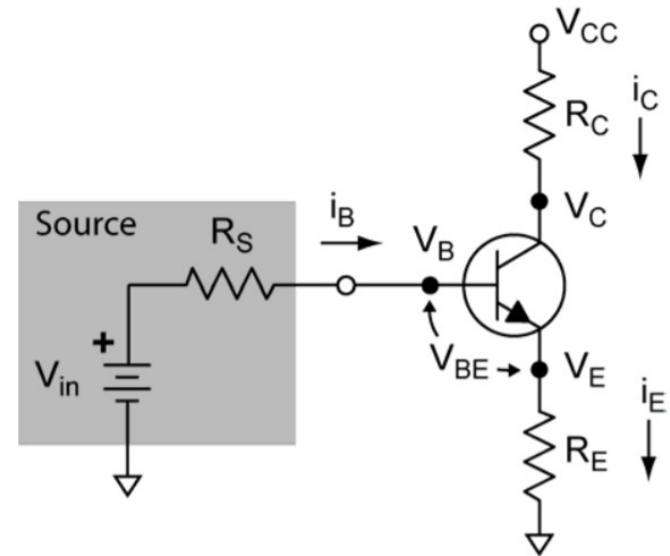
$$I_{out} = I_E$$

$$I_B = \frac{I_E}{\beta + 1}$$

$$V_{out} = V_{in} - \left(\frac{I_E}{\beta + 1} \right) R_S - V_{BE} = V_{in} - \left(\frac{I_{out}}{\beta + 1} \right) R_S - V_{BE} = -I_{out} \left(\frac{R_S}{\beta + 1} \right) + (V_{in} - V_{BE})$$

$$Z_{out} = -\frac{dV_{out}}{dI_{out}} = -\frac{d}{dI_{out}} \left(-I_{out} \left(\frac{R_S}{\beta + 1} \right) + (V_{in} - V_{BE}) \right) = \left(\frac{R_S}{\beta + 1} \right)$$

$$Z_{out} = \left(\frac{R_S}{\beta + 1} \right) \approx \frac{R_S}{\beta}$$

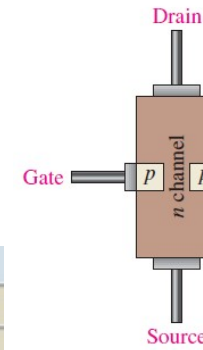


FET (Field Effect Transistor)

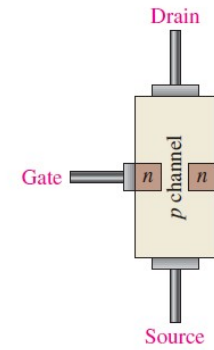
- Junction field-effect transistor (JFET)
- Metal oxide semiconductor field-effect transistor (MOSFET)

Terminal İsmi (Kısaltma)	BJT Transistördeki karşılığı
Drain (D)	Collector (C)
Gate (G)	Base (B)
Source (S)	Emmitter (E)

Field Effect Transistor (FET)	Bipolar Junction Transistor (BJT)
Düşük voltaj kazancı	Yüksek voltaj kazancı
Yüksek akım kazancı	Düşük akım kazancı
Çok yüksek giriş empedansı	Düşük giriş empedansı
Yüksek çıkış empedansı	Düşük çıkış empedansı
Düşük gürültü seviyesi	Daha yüksek gürültü seviyesi
Hızlı anahtarlama zamanı	Daha yavaş anahtarlama zamanı
Statik elektrikten kolay etkilenme	Statik elektrığe karşı dayanıklı
Voltaj kontrollü	Akım kontrollü
Daha pahalı	Ucuz

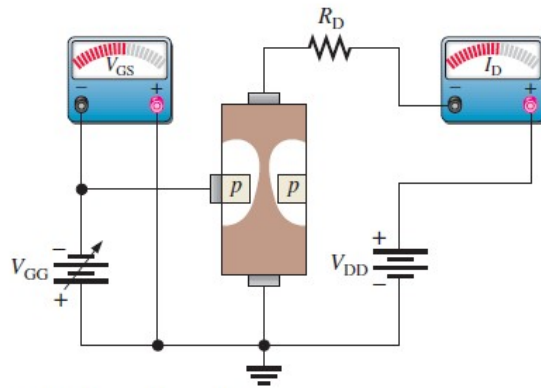


(a) n channel

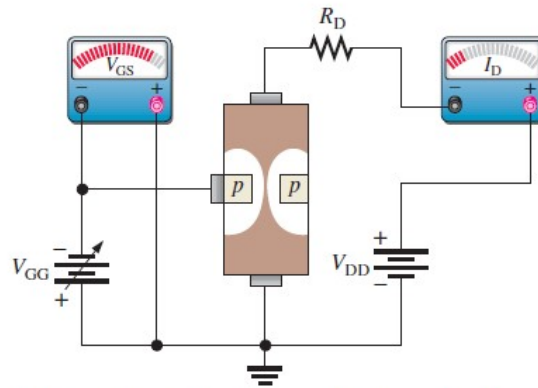


(b) p channel

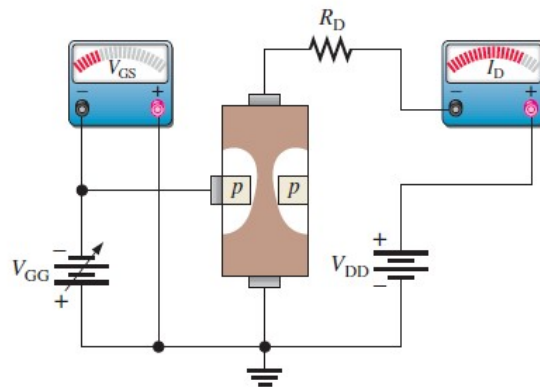
Junction field-effect transistor (JFET)



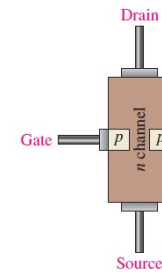
(a) JFET biased for conduction



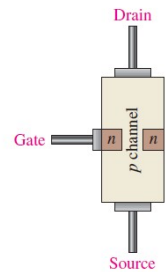
(b) Greater V_{GG} narrows the channel (between the white areas) which increases the resistance of the channel and decreases I_D .



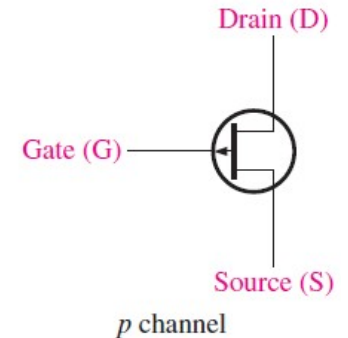
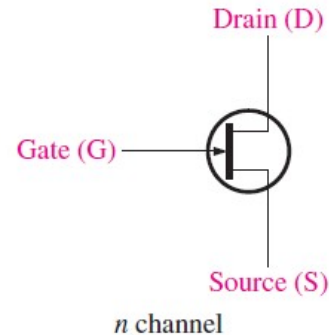
(c) Less V_{GG} widens the channel (between the white areas) which decreases the resistance of the channel and increases I_D .



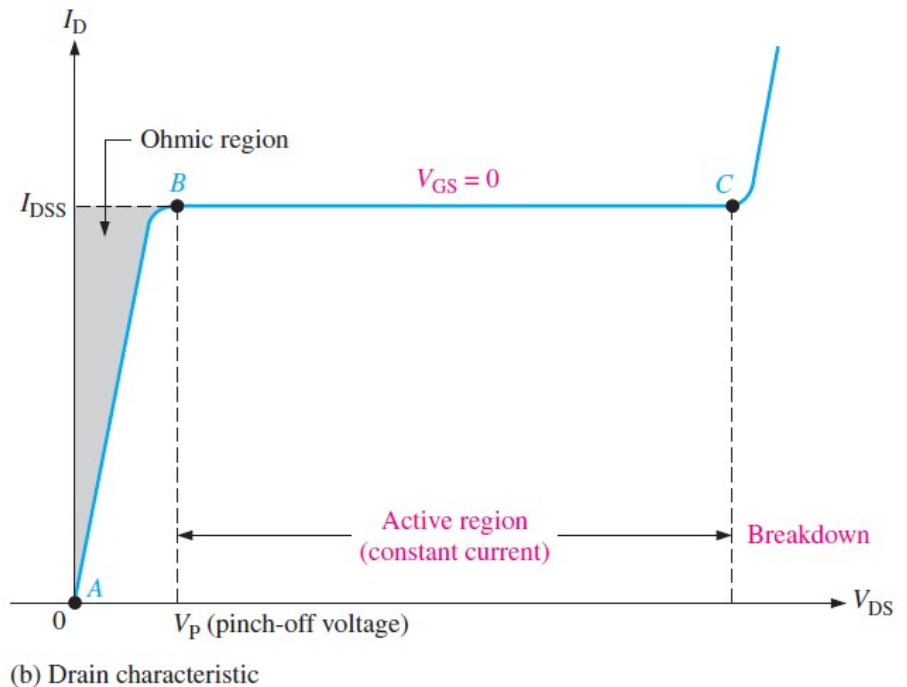
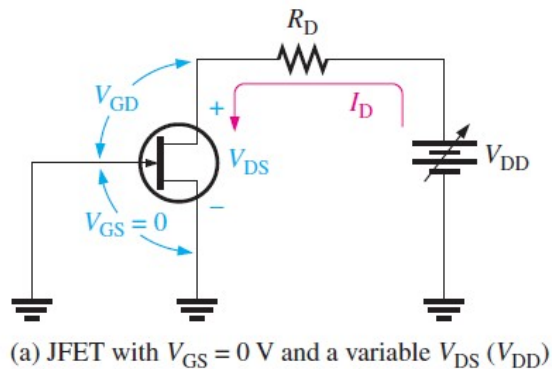
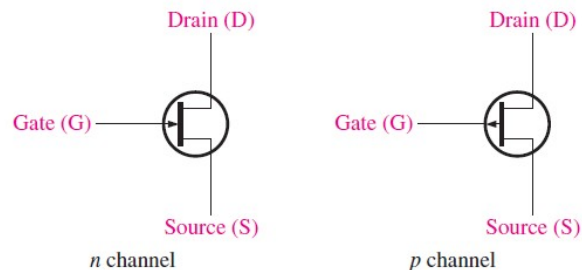
(a) n channel

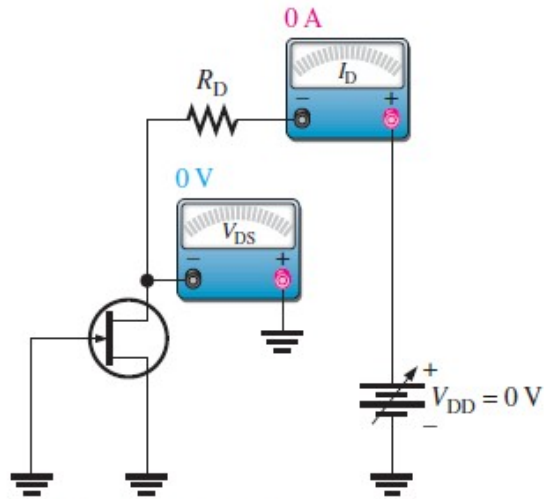


(b) p channel

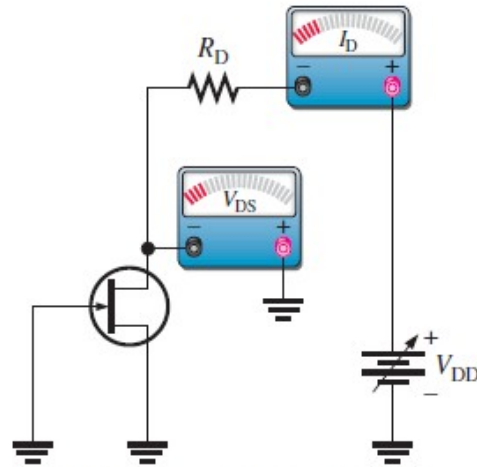


Junction field-effect transistor (JFET)

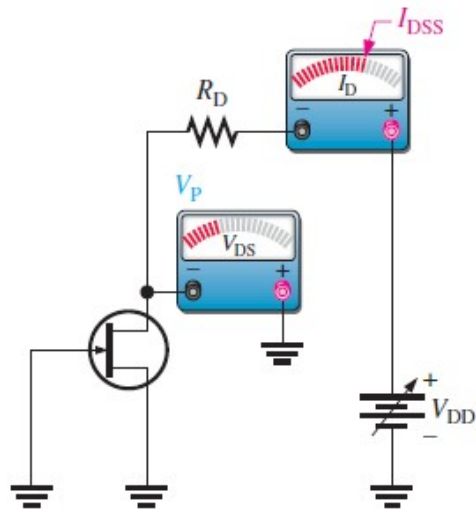




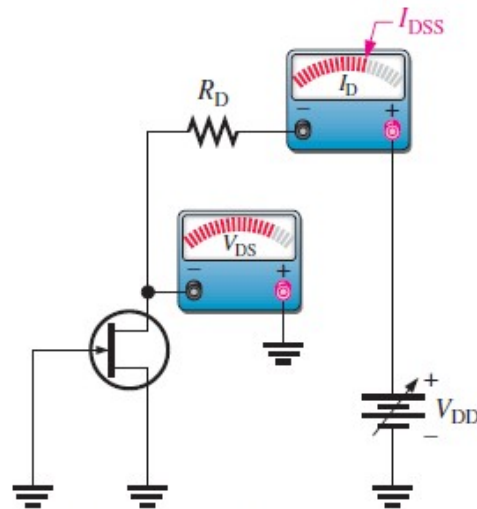
(a) When $V_{DS} = 0$, $I_D = 0$.



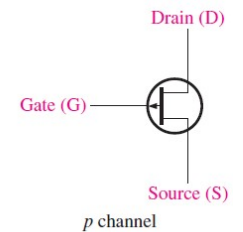
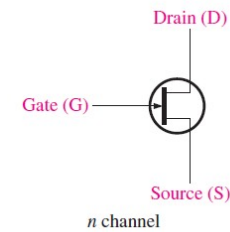
(b) I_D increases proportionally with V_{DS} in the ohmic region.

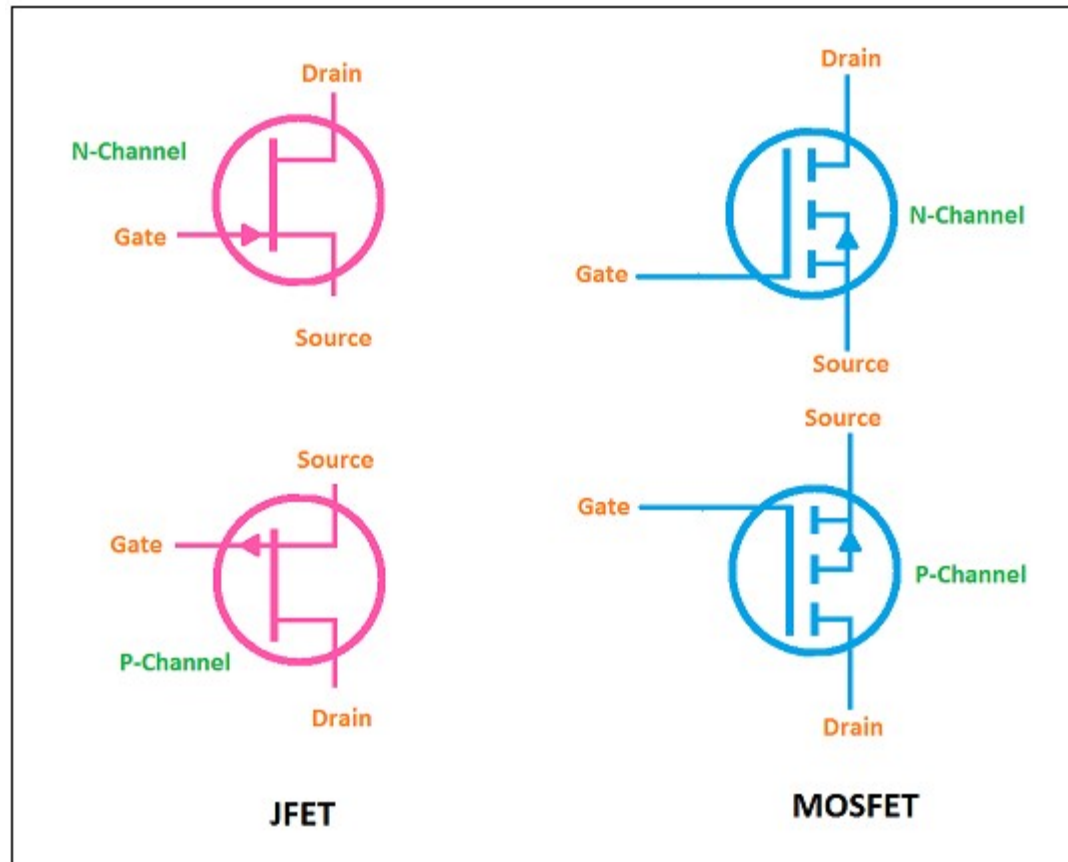


(c) When $V_{DS} = V_P$, I_D is constant and equal to I_{DSS} .



(d) As V_{DS} increases further, I_D remains at I_{DSS} until breakdown occurs.



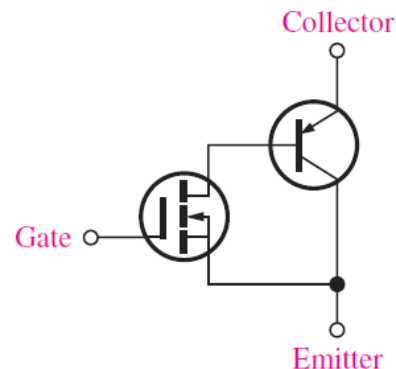
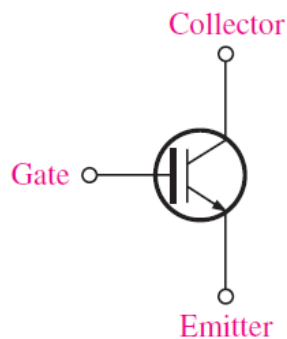


<https://www.etechnog.com/2019/07/mosfet-vs-jfet-differences.html>

IGBT (insulated-gate bipolar transistor)

- The **IGBT** is a device that has the output conduction characteristics of a BJT but is voltage controlled like a MOSFET; it is an excellent choice for many high-voltage switching applications.

FEATURES	IGBT	MOSFET	BJT
Type of input drive	Voltage	Voltage	Current
Input resistance	High	High	Low
Operating frequency	Medium	High	Low
Switching speed	Medium	Fast (ns)	Slow (μs)
Saturation voltage	Low	High	Low



BJT

output current is a function of current signal into base



a current signal into base modulates the number of current carriers available to carry current thru device

FET

output current is a function of voltage signal at gate



a voltage signal at gate modulates channel resistance, and thus current thru device

IGBT

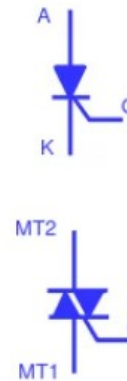
output current is a function of voltage signal at gate



a voltage signal at gate modulates both channel resistance and available current carriers, and thus current thru device

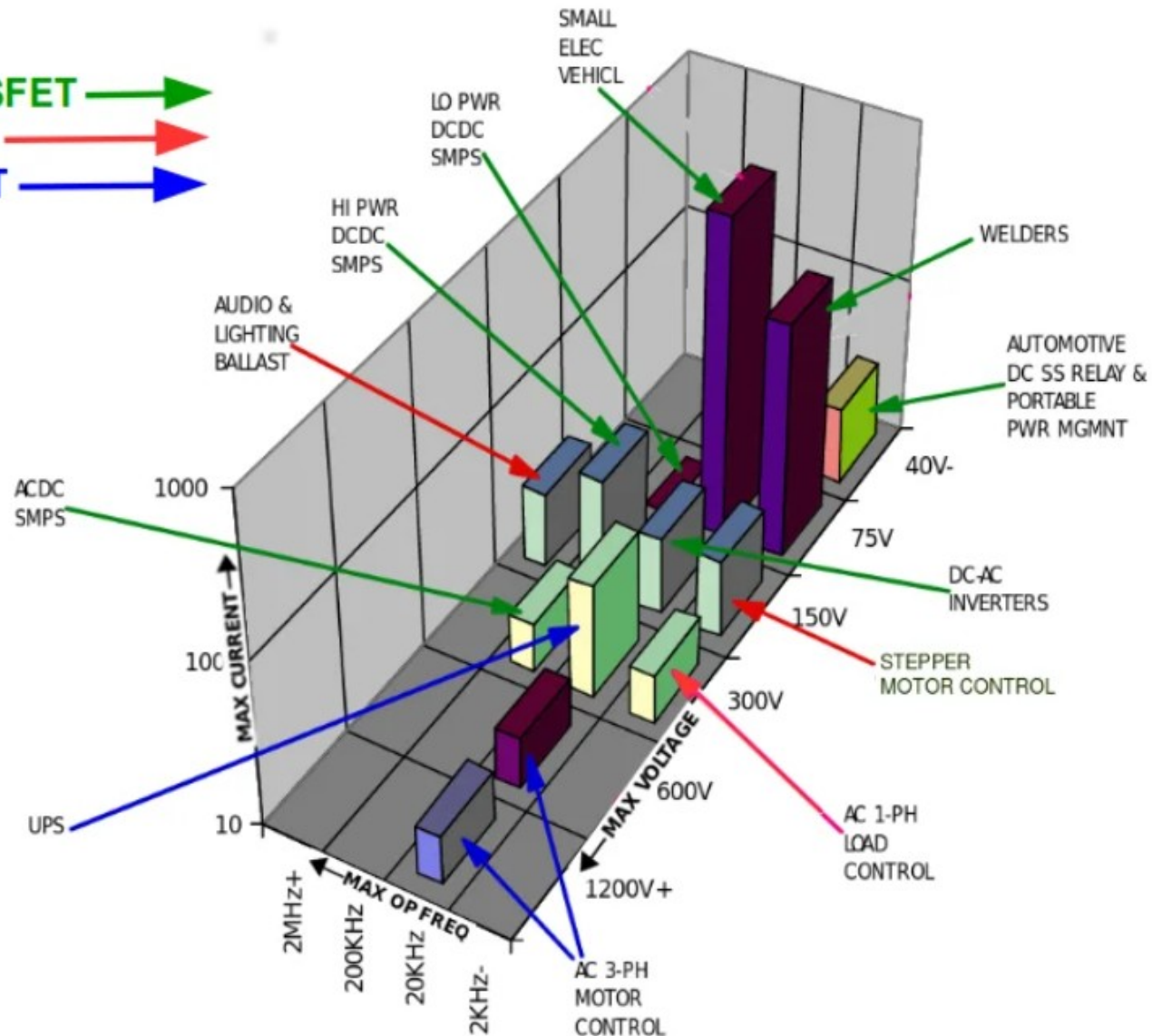
Thyristor

output latches *on* when triggered by pulse signal at gate



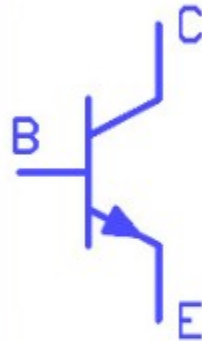
a small gate pulse triggers the device on. It stays latched on until the current thru the device drops below its holding current (I_H)

MOSFET →
BJT →
IGBT →



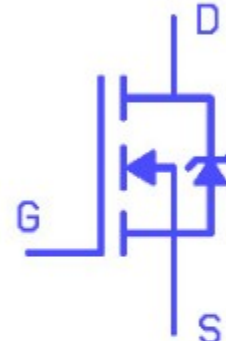
<https://www.allaboutcircuits.com/technical-articles/fet-vs-bjt-vs-igbt-whats-the-right-choice-for-your-power-stage-design/>

BJT:
HIGH VOLTAGE
LOW CURRENT



- Low ON losses due to relatively low $V_{CE(sat)}$
- Moderate switching speeds (up to 100 kHz)
- Currents limited to a few Amps at high volts
- Requires complex base drive circuitry, feedback

MOSFET:
HIGH CURRENT
LOW VOLTAGE



- Low $R_{DS(on)}$ of low voltage MOSFET can carry high currents
- High switching speeds (up to 500 kHz)
- Simple gate drive circuitry



IGBT:
HIGH VOLTAGE
HIGH CURRENT

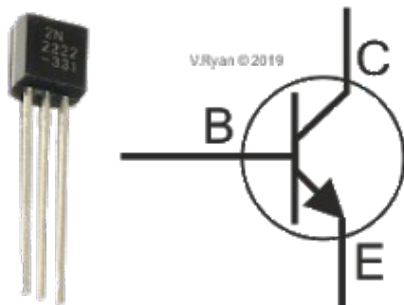


- Much Lower ON losses than High Voltage MOSFET
- Low switching speeds (<50 kHz)
- Simple gate drive circuitry

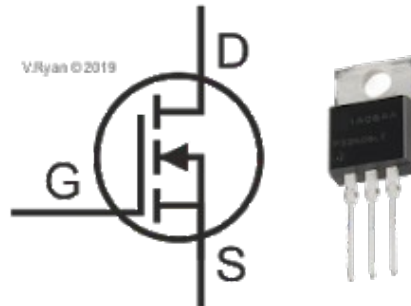
BJT vs. FET

BJTs vs FETs		
	BJTs	FETs
How it operates	BJTs are current-controlled. They require a biasing current to the base terminal for operation.	FETs are voltage-controlled. They only require voltage applied to the gate to turn the FET either on or off. They do not require a biasing current for operation.
Input Impedance	BJTs offer smaller input impedances, meaning they draw more current from the power circuit feeding it, which can cause loading of the circuit.	FETs offer greater input impedance than BJTs. This means that they practically draw no current and therefore do not load down the power circuit that's feeding it.
Gain (Transconductance)	BJTs offer greater gain at the output than FETs.	The gain (or transconductance) of FETs are smaller than for BJTs.
Size	BJTs are larger in size and therefore take up more physical space than FETs normally.	FETs can be manufactured much smaller than BJTs. This is especially important for integrated circuits that are composed up of many transistors.
Popularity	BJTs are less popular and less widely used	FETs are definitely more popular and widely used in commercial circuits today than BJTs
Cost	BJTs are cheaper to manufacture	FETs, especially MOSFETs, are more expensive to manufacture

BJT
(bipolar junction transistor)

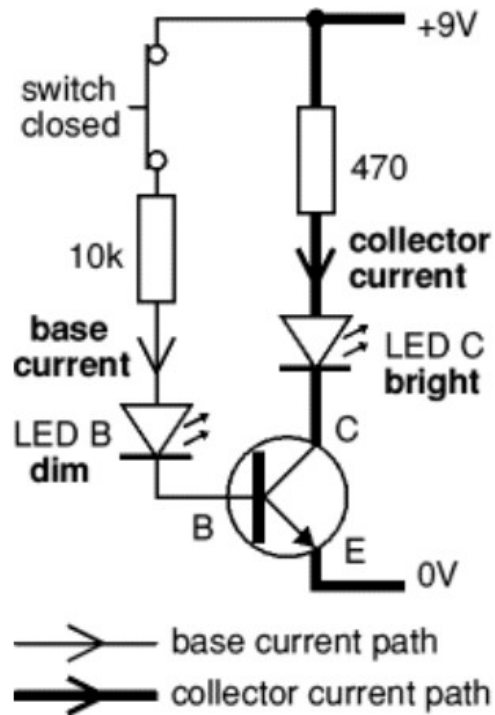


FET
(Field-Effect Transistors)



Transistor Circuits

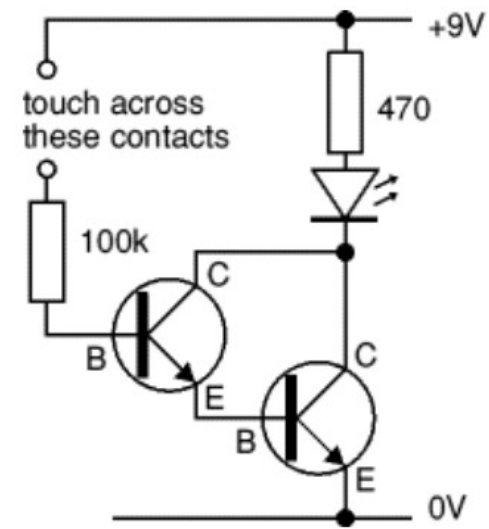
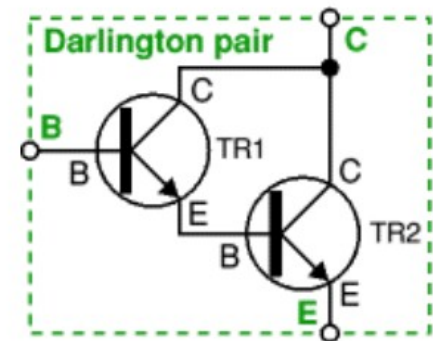
- The small **base current** controls the larger **collector current**.



BC108, BC182 or BC548

<https://dlb.sa.edu.au/rehsmoodle/file.php/466/kpsec.freeuk.com/trancirc.htm>

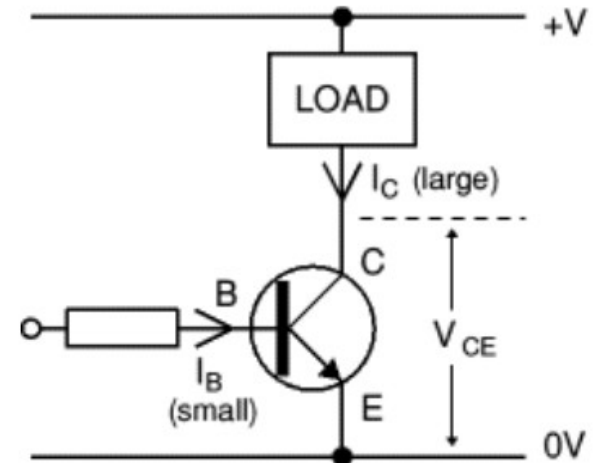
- **Darlington pair current gain, $h_{FE} = h_{FE1} \times h_{FE2}$**
- This gives the Darlington pair a very high current gain, such as 10000, so that only a tiny base current is required to make the pair switch on.
- A Darlington pair is sufficiently sensitive to respond to the small current passed by your skin and it can be used to make a touch-switch as shown in the diagram. For this circuit which just lights an LED the two transistors can be any general purpose low power transistors. The 100kohm resistor protects the transistors if the contacts are linked with a piece of wire.



Touch switch circuit

<https://dlb.sa.edu.au/rehsmoodle/file.php/466/kpsec.freeuk.com/trancirc.htm>

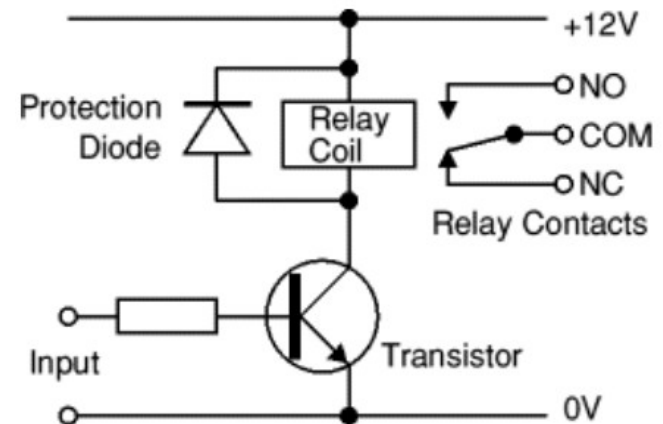
- The power developed in a switching transistor is very small:
- In the **OFF** state: power = $I_C \times V_{CE}$, but $I_C = 0$, so the power is zero.
- In the **full ON** state: power = $I_C \times V_{CE}$, but $V_{CE} = 0$ (almost), so the power is very small.
- This means that the transistor should not become hot in use and you do not need to consider its maximum power rating.



<https://dlb.sa.edu.au/rehsmoodle/file.php/466/kpsec.freeuk.com/trancirc.htm>

- If the load is a **motor**, **relay** or **solenoid** (or any other device with a coil) a diode must be connected across the load to protect the transistor (and chip) from damage when the load is switched off. The diagram shows how this is connected 'backwards' so that it will normally NOT conduct. Conduction only occurs when the load is switched off, at this moment current tries to continue flowing through the coil and it is harmlessly diverted through the diode. Without the diode no current could flow and the coil would produce a damaging high voltage 'spike' in its attempt to keep the current flowing.

<https://dlb.sa.edu.au/rehsmoodle/file.php/466/kpsec.freeuk.com/trancirc.htm>



Relay vs Transistor

Advantages of relays:

- Relays can switch **AC and DC**, transistors can only switch DC.
- Relays can switch **high voltages**, transistors cannot.
- Relays are a better choice for switching **large currents** ($> 5A$).
- Relays can switch **many contacts** at once.

Disadvantages of relays:

- Relays are **bulkier** than transistors for switching small currents.
- Relays **cannot switch rapidly**, transistors can switch many times per second.
- Relays **use more power** due to the current flowing through their coil.
- Relays **require more current than many chips can provide**, so a low power transistor may be needed to switch the current for the relay's coil.

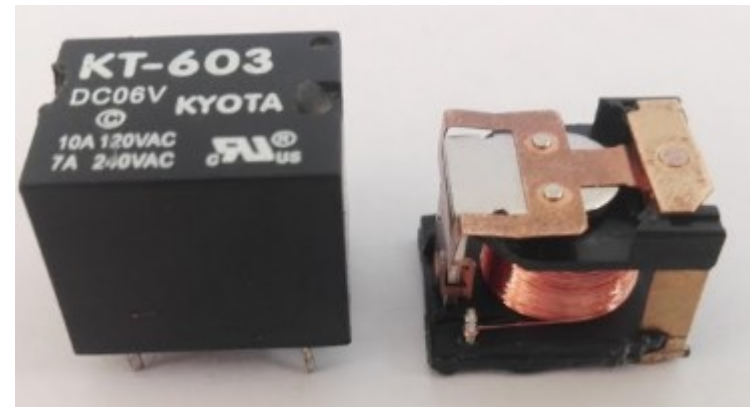
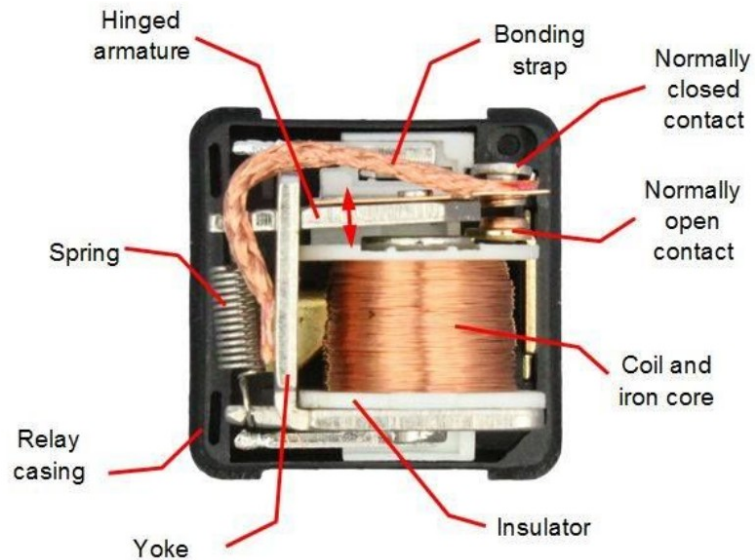


Relays

Photographs © [Rapid Electronics](#)

<https://dlb.sa.edu.au/rehsmoodle/file.php/466/kpsec.freeuk.com/trancirc.htm>

Relay



<https://i.pinimg.com/originals/2a/19/12/2a1912243e58ffd61f348d5da995fd0e.jpg>

Switching Transistor

The procedure below explains how to choose a suitable switching transistor.

1. The transistor's maximum collector current $I_{C(max)}$ must be greater than the load current I_C .

$$\text{load current } I_C = \frac{\text{supply voltage } V_s}{\text{load resistance } R_L}$$

2. The transistor's minimum current gain $h_{FE(min)}$ must be at least **five** times the load current I_C divided by the maximum output current from the chip.

$$h_{FE(min)} > 5 \times \frac{\text{load current } I_C}{\text{max. chip current}}$$

3. Choose a transistor which meets these requirements and make a note of its properties: $I_{C(max)}$ and $h_{FE(min)}$.

There is a table showing technical data for some popular transistors on the [transistors](#) page.

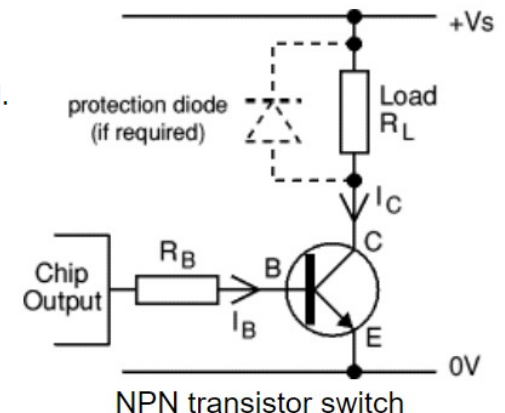
4. Calculate an approximate value for the base resistor:

$$R_B = \frac{V_C \times h_{FE}}{5 \times I_C} \quad \text{where } V_C = \text{chip supply voltage} \\ \text{(in a simple circuit with one supply this is } V_s)$$

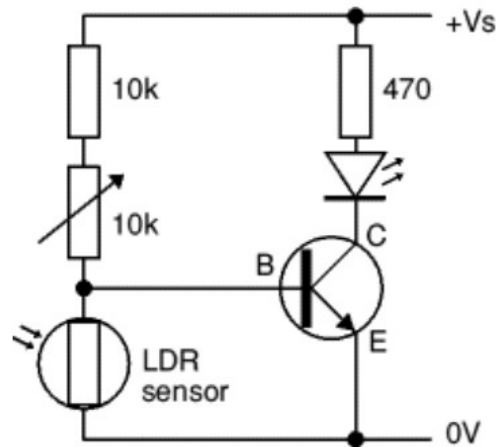
For a simple circuit where the chip and the load share the same power supply ($V_C = V_s$) you may prefer to use: $R_B = 0.2 \times R_L \times h_{FE}$

Then choose the nearest standard value for the base resistor.

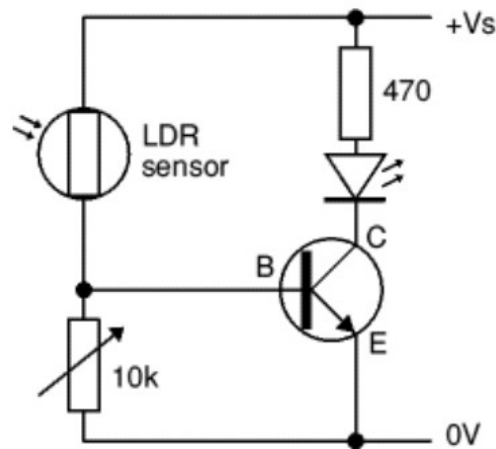
5. Finally, remember that if the load is a motor or relay coil a **protection diode** is required.

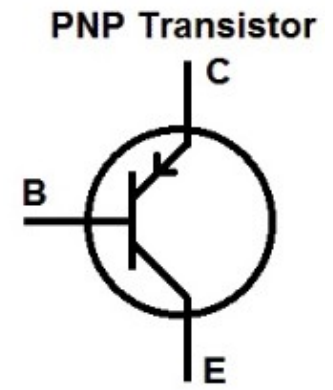
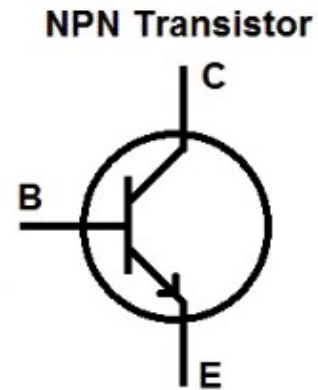
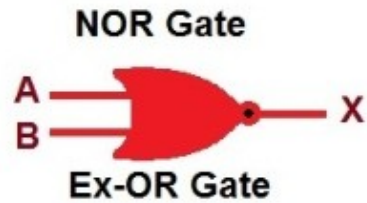
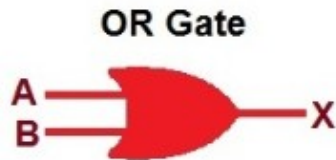
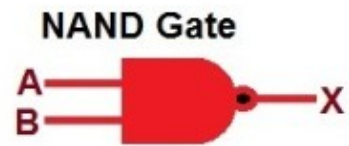


Using a transistor switch with sensors

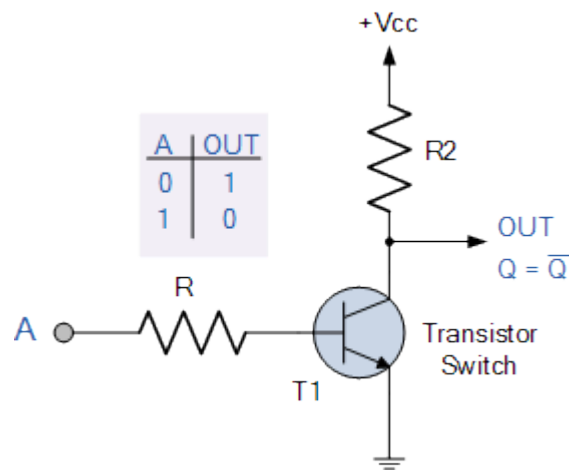
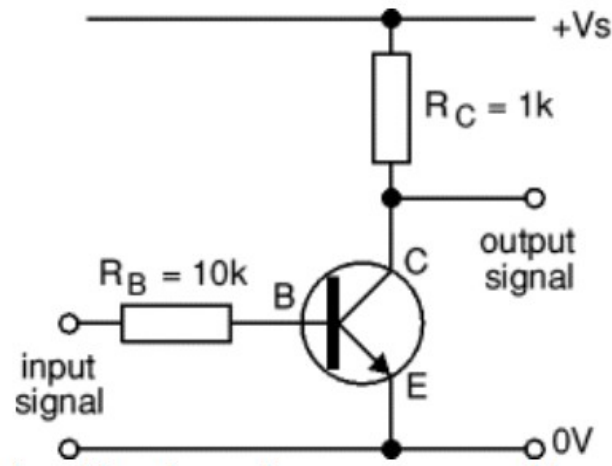


LED lights when the LDR is **dark**

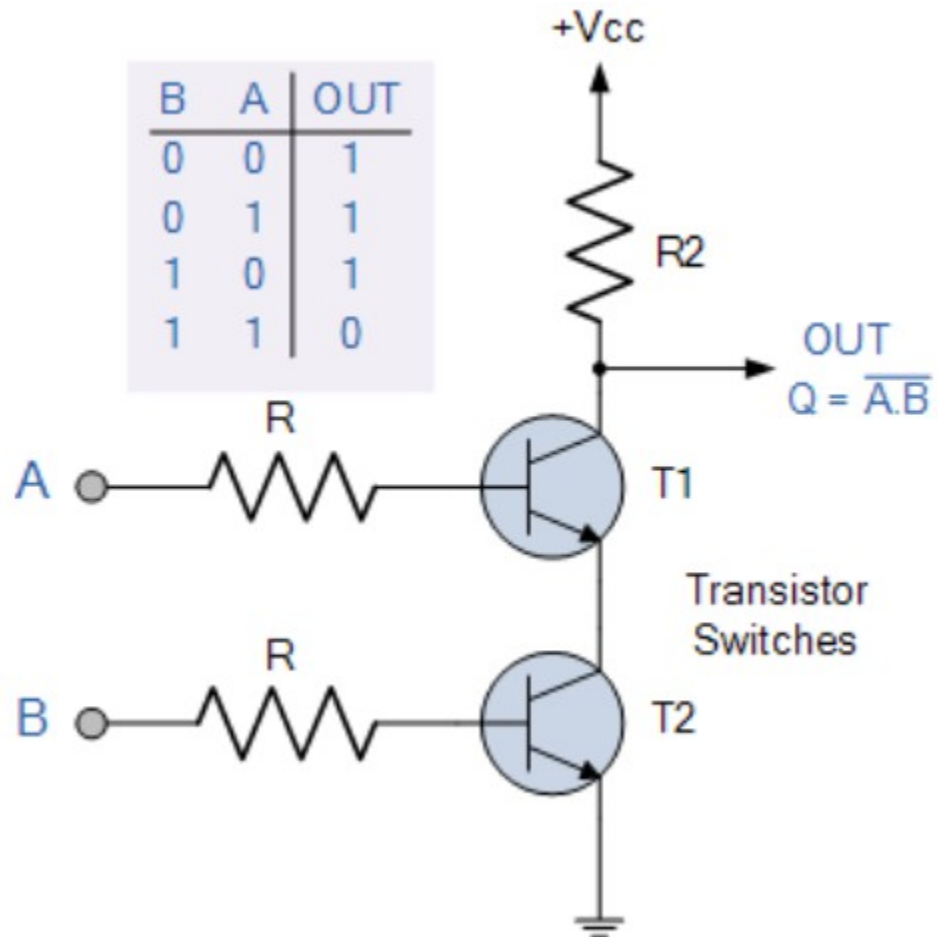




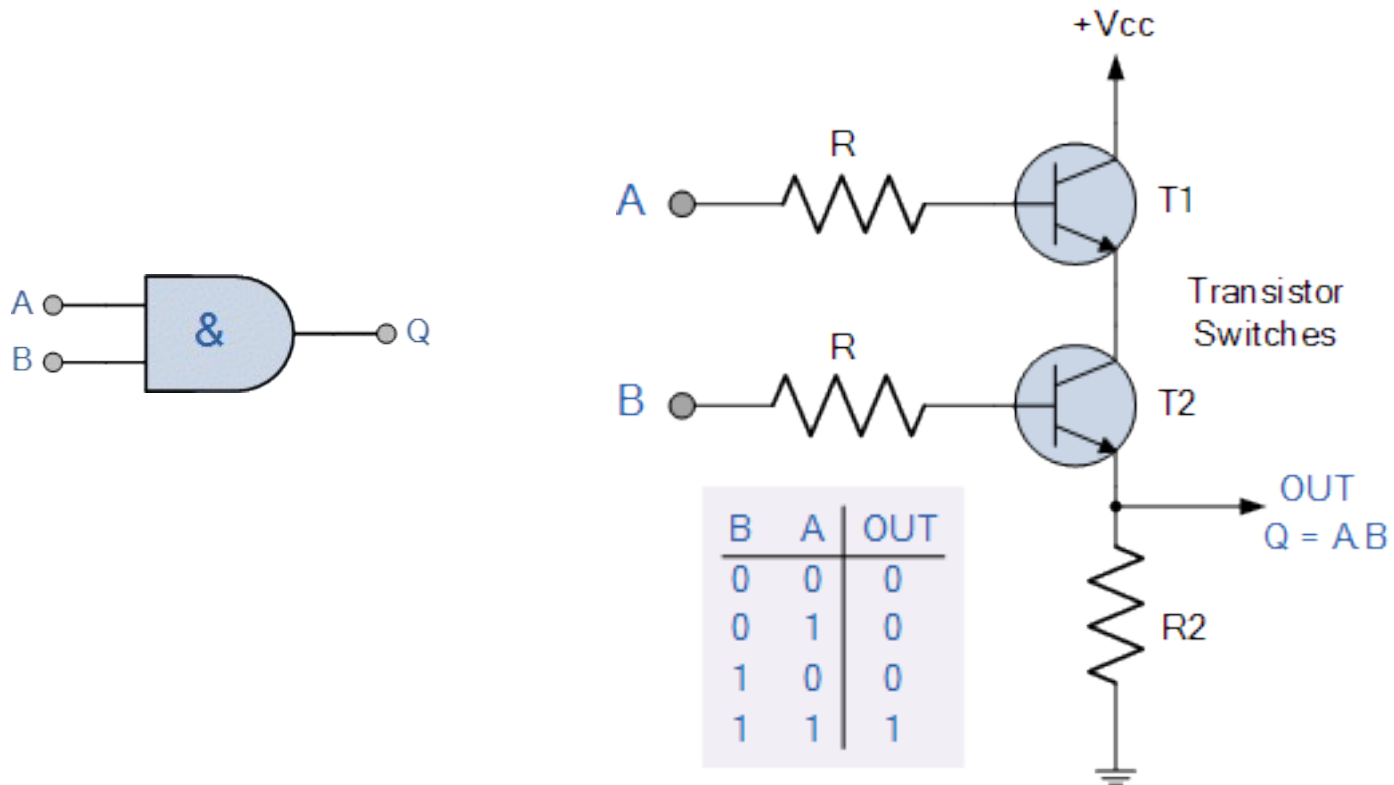
A transistor inverter (NOT gate)



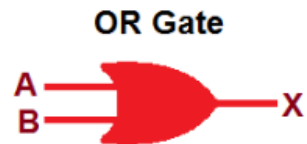
NAND Gate



AND Gate

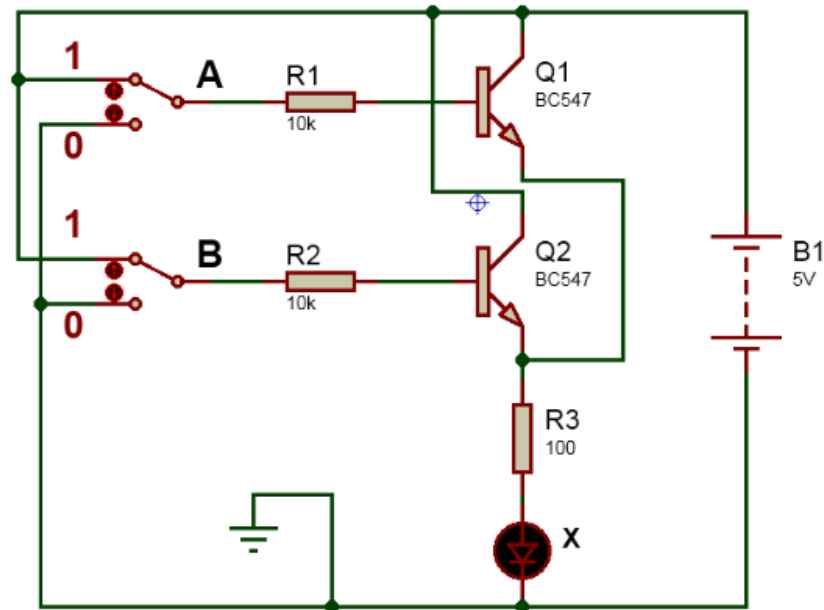


OR Gate



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

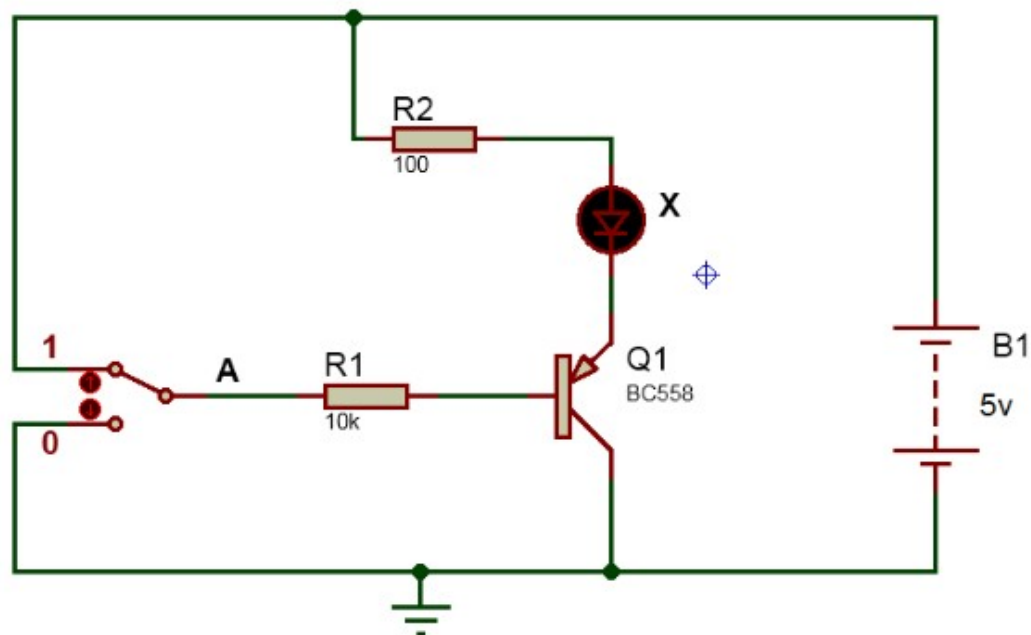


NOT Gate



Truth Table

A	X
0	1
1	0

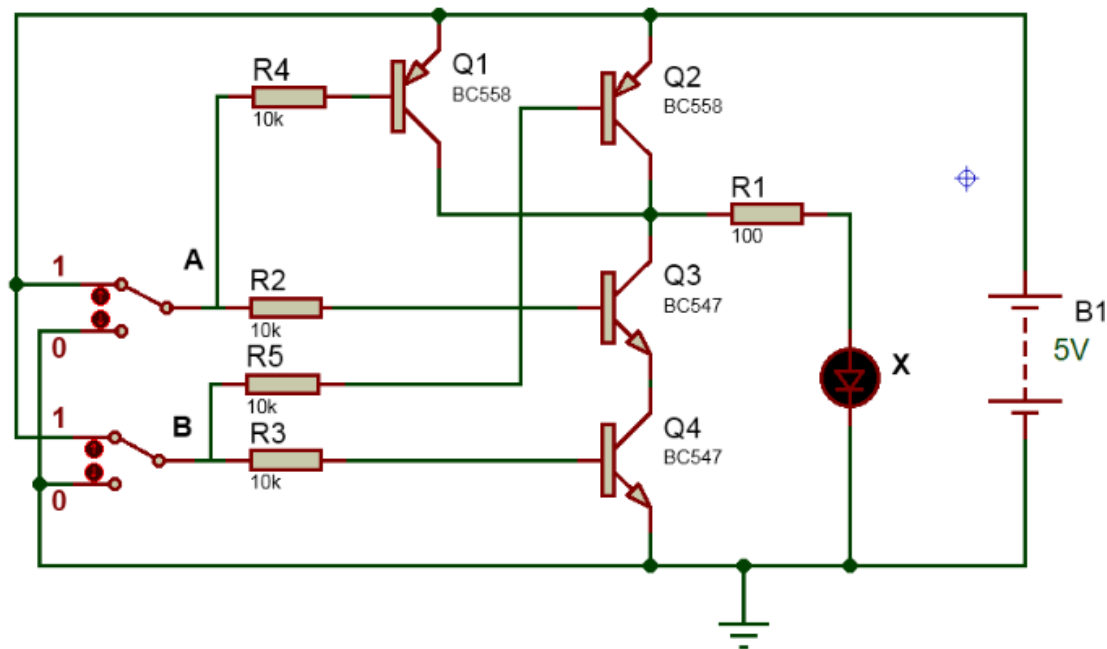


NAND Gate



Truth Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

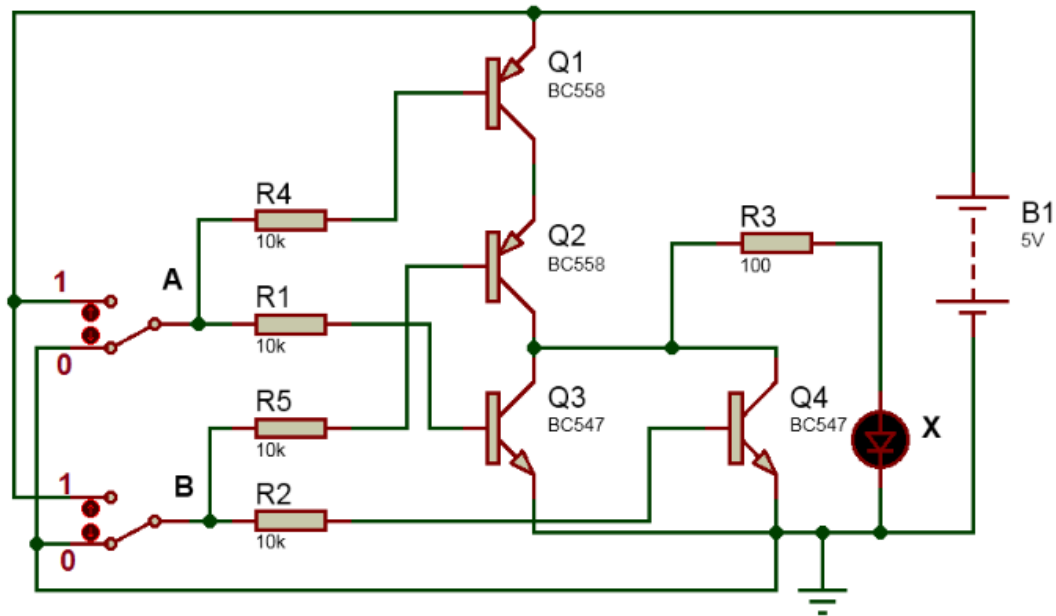


NOR Gate



Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

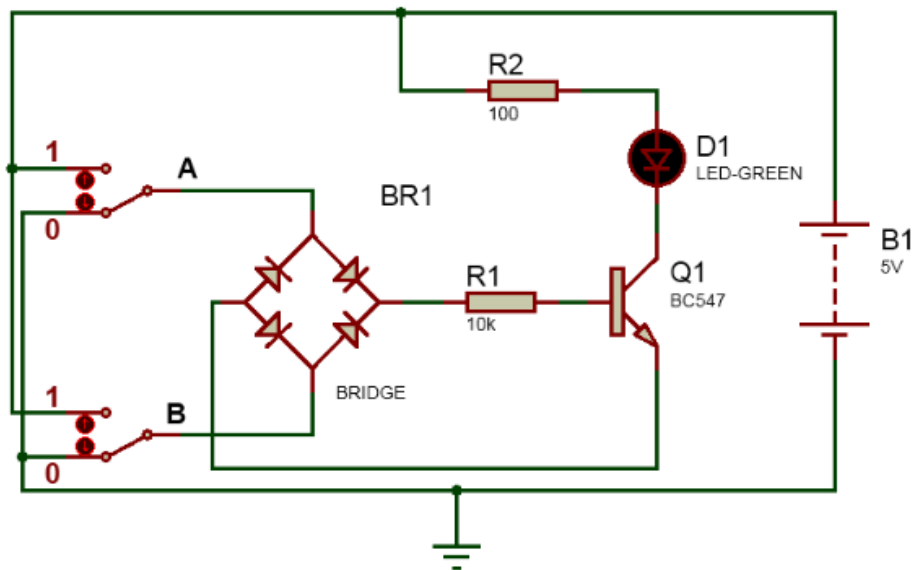


Ex-OR Gate



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



Ex-NOR Gate

Ex-NOR Gate



Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

- > when switch **A** is press **0** and switch **B** is press **0**, the **LED** is **ON**
- > when switch **A** is press **0** and switch **B** is press **1**, the **LED** is **OFF**
- > when switch **A** is press **1** and switch **B** is press **0**, the **LED** is **OFF**
- > when switch **A** is press **1** and switch **B** is press **1**, the **LED** is **ON**

